



Department of Electrical Engineering and Computer Science

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

6.1810 Fall 2024

Quiz I

All problems are open-ended questions. In order to receive credit you must answer the question as precisely as possible. You have 80 minutes to finish this quiz.

Write your name on this cover sheet AND at the bottom of each page of this booklet.

Some questions may be harder than others. Read them all through first and attack them in the order that allows you to make the most progress. If you find a question ambiguous, be sure to write down any assumptions you make. Be neat. If we can't understand your answer, we can't give you credit!

THIS EXAM IS OPEN BOOK AND OPEN LAPTOP, but CLOSED NETWORK.

Please do not write in the boxes below.

I (xx/8)	II (xx/10)	III (xx/10)	IV (xx/10)	V (xx/15)	VI (xx/5)	VII (xx/3)	(xx/61)
3	10	10	5	15	0	3	46

Gradescope E-Mail Address:

Name:

F. He

I System Calls

Suppose that you run the following program on xv6. (See Chapter 1 of the xv6 book for a description of the system calls.)

```
int main() {  
    if(fork() == 0){  
        write(1, "x", 1);  
    }  
    if(fork() == 0){  
        write(1, "y", 1);  
    }  
    wait(0);  
    wait(0);  
}
```

None of the system calls fail.

1. [3 points]: Which of the following outputs could this program produce? Circle all that apply.

A. xy

B. yx

C. xyy

D. yxy

E. yyx

Name:

J. H.

Now consider running this xv6 program:

```
char aaa = 'w';  
char bbb;
```

```
int
```

```
main() {  
    int fds[2];
```

1

```
    aaa = 'x';
```

```
    pipe(fds);
```

```
    if(fork() == 0){
```

2

```
        write(fds[1], &aaa, 1);
```

3

```
        bbb = 'y';
```

```
        while(1){
```

```
            sleep(1);
```

```
        }
```

```
    } else {
```

4

```
        aaa = 'z';
```

5

```
        read(fds[0], &bbb, 1);
```

6

```
        write(1, &bbb, 1);
```

```
    }
```

No system calls fail.

2. [5 points]: What are the possible outputs? Circle all that apply.

A. Nothing

B. x

C. y

D. z

1 2 4 5 6 3

1 2 4 5 3 6

1 4 2 5 6 3

B.
fork py
mem

Name:

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II Page tables

Fig 1 shows how RISC-V translates virtual addresses to physical addresses and the format of a Page Table Entry (PTE).

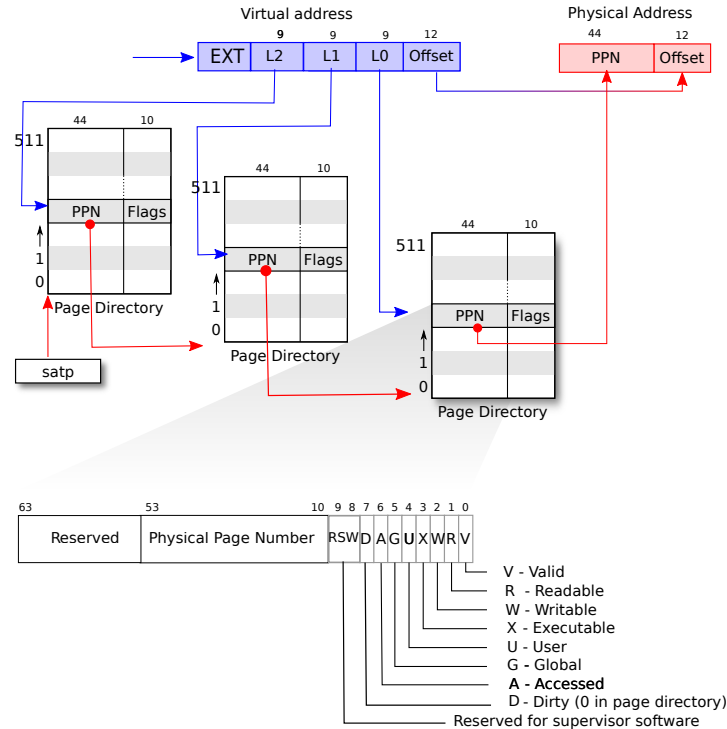


Figure 1: RISC-V address translation details.

3. [5 points]: For the virtual address $0xFFFFD000$ what is the index into the L0 page directory? Please write down a hex number.

$$9 \text{ of } FFD = 0x1FD$$

4. [5 points]: Ben creates a page table that has PTE $0x21FD10D7$ for virtual address 0 and PTE $0x21FD10D6$ for virtual address 4096. If Ben stores the value 1 to virtual address 0 and then loads from virtual address 4096, does the load return the value 1? Circle the one best answer.

A. Yes

B. No

DAUWRV

DAUWR (NOT VALID)

Page fault
Name:

III Superpages

Ben observes that KERNBASE is 0x80000000 and PHYSTOP is 0x88000000 in the two calls to kvmmap shown below on line 20 and 23.

```

1 // Make a direct-map page table for the kernel.
2 pagetable_t
3 kvmmake(void)
4 {
5     pagetable_t kpgtbl;
6
7     kpgtbl = (pagetable_t) kalloc();
8     memset(kpgtbl, 0, PGSIZE);
9
10    // uart registers
11    kvmmap(kpgtbl, UART0, UART0, PGSIZE, PTE_R | PTE_W);
12
13    // virtio mmio disk interface
14    kvmmap(kpgtbl, VIRTIO0, VIRTIO0, PGSIZE, PTE_R | PTE_W);
15
16    // PLIC
17    kvmmap(kpgtbl, PLIC, PLIC, 0x4000000, PTE_R | PTE_W);
18
19    // map kernel text executable and read-only.
20    kvmmap(kpgtbl, KERNBASE, KERNBASE, (uint64)etext-KERNBASE, PTE_R | PTE_X);
21
22    // map kernel data and the physical RAM we'll make use of.
23    kvmmap(kpgtbl, (uint64)etext, (uint64)etext, PHYSTOP-(uint64)etext, PTE_R | PTE_W);
24
25    // map the trampoline for trap entry/exit to
26    // the highest virtual address in the kernel.
27    kvmmap(kpgtbl, TRAMPOLINE, (uint64)trampoline, PGSIZE, PTE_R | PTE_X);
28
29    // allocate and map a kernel stack for each process.
30    proc_mapstacks(kpgtbl);
31
32    return kpgtbl;
33 }

```

5. [5 points]: How many physical pages of memory do these two kvmmap calls allocate for page directories to map this region of memory? Circle the one best answer.

Handwritten notes and diagram illustrating the calculation of physical pages for the two `kvmmap` calls.

Handwritten notes:

- $0 - 40000000$ (with an arrow pointing to the first `kvmmap` call)
- 2^{27} (written above the first `kvmmap` call)
- $1 + 64$ (written next to the first `kvmmap` call)
- $80000000 - 00000000$ (written next to the second `kvmmap` call)
- 2^{27} (written next to the second `kvmmap` call)
- $2^{27} / 2^{21} = 2^6 = 64$ (written at the bottom)

Diagram illustrating the mapping of memory regions:

- A box labeled "Already allocated" with a checkmark and the number "1" below it.
- A box labeled "Occupy 1 page in total" with a checkmark and the number "5" below it.
- A box labeled "64 of the PTE pages" with a checkmark and the number "64" below it.

Multiple choice options:

- A. 65
- B. 32768

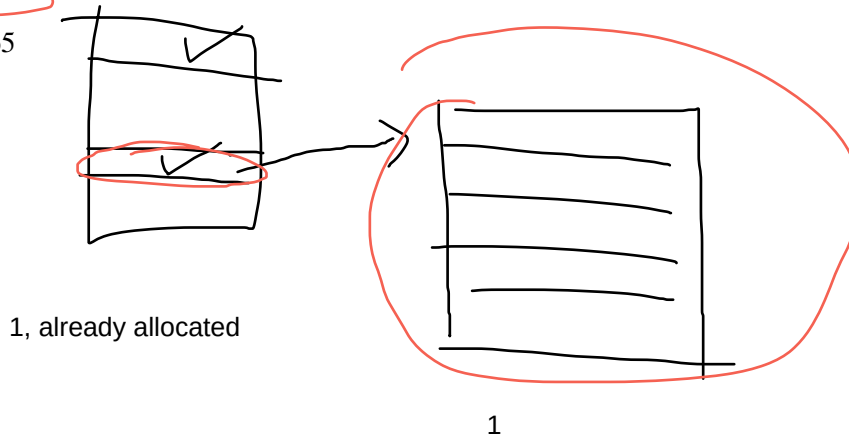
Handwritten signature: "S. J. L."

Eve observes that that these two calls map physical memory contiguously and modifies xv6 to use super-pages for this region of memory.

6. [5 points]: How many physical pages of memory does Eve's modified xv6 allocate for page directories to map this memory? Circle the one best answer.

A. 1

B. 65



Name:

J. H.

IV Traps

xv6 user code uses the RISC-V `ecall` instruction to start a system call. Among other things, `ecall` jumps to the start of the trampoline code.

Ben wonders whether it would work to replace `ecall` with `jal 0x3fffffff000`, a RISC-V subroutine call instruction that jumps to the start of the trampoline. (In fact, Ben would need a few more instructions to load the 64-bit `0x3fffffff000` into a register, followed by `jalr`.)

7. [5 points]: Why won't Ben's idea work? Circle all that apply.

- A. `ecall` is needed because `ecall` changes `satp` to point to the kernel page table.
- B. `ecall` is needed because `ecall` saves the 32 user registers.
- C. `ecall` is needed because the trampoline's address has no PTE in the user page table.
- D. `ecall` is needed because the address of the trampoline page is different for different processes.
- E. none of the above.

1. Trampoline is not mapped with U.
Only `ecall` that switches to S first can jump to there.
2. `ecall` saves the previous program counter for `sret`.

Name:

CH

When a device interrupt occurs while user code is executing, `usertrap()` in `kernel/trap.c` saves the RISC-V `sepc` register, which holds the program counter value at which the interrupt occurred:

```
// save user program counter.
p->trapframe->epc = r_sepc();
```

When returning to user space, `usertrapret()` copies that saved value back to `sepc`:

```
// set S Exception Program Counter to the saved user pc.
w_sepc(p->trapframe->epc);
```

In many cases the value in the `sepc` register doesn't change during interrupt handling, so that this save and restore doesn't actually change the value in `sepc`.

8. [5 points]: There is a situation in which, after a process traps into the kernel due to a device interrupt, the `sepc` register will have changed by the time of that process's return via `usertrapret()`. Please briefly explain what that situation is.

When a trap occurred between `intr_on()` and `usertrapret()`

Per intr $\rightarrow$ clock intr $\rightarrow$ yield
 $\swarrow$
 good luck

Name:

8

V Threads

`scheduler()` in `kernel/proc.c` context-switches with this call:

```
swtch(&c->context, &p->context);
```

Ben accidentally changes `c` to `p`:

```
swtch(&p->context, &p->context);
```

so that now (ignoring comments) the `scheduler()` loop looks like this:

```
for(;;) {  
    intr_on();  
    for(p = &proc[0]; p < &proc[NPROC]; p++) {  
        acquire(&p->lock);  
        if(p->state == RUNNABLE) {  
            p->state = RUNNING;  
            c->proc = p;  
            swtch(&p->context, &p->context);  
            c->proc = 0;  
        }  
        release(&p->lock);  
    }  
}
```

Save regs to $p \rightarrow ctx$
read regs from $p \rightarrow ctx$
= nothing

What will go wrong when Ben runs `xv6`? Circle the one best answer.

9. [5 points]:

- A. `scheduler()`'s call to `swtch()` will context-switch to a different kernel thread, but that thread will soon crash or panic because the register contents are incorrect.
- ☒ B. `scheduler()` will loop forever without ever context-switching to a different kernel thread.
- C. `swtch()` will context-switch to a different kernel thread, which will execute correctly until it tries to `swtch()` back to the scheduler thread, at which point the fact that `c->context` was never initialized will cause a crash or panic.
- D. `scheduler()`'s call to `swtch()` will run the process `p` twice, instead of just once, giving it an unfair advantage.
- E. `scheduler()` will likely `swtch()` to a kernel thread that is already running on a different CPU.

Name:

GH

An xv6 process that is giving up the CPU calls `sched()`, which calls `swtch()`. Alyssa wants to know what address that `swtch()` call will return to. She would like to add a `printf()` just before the `swtch()` in `sched()` to print that address, so that `sched()` looks like this:

```
printf("%p\n", (void *) ???);  
swtch(&p->context, &c->context);
```

10. [5 points]: What value should Alyssa print in order to find out where `sched()`'s call to `swtch()` will return? Circle the one best answer.

A. `p->context.ra`

B. `&scheduler`

~~C. `r_sepc()`~~

D. `c->context.ra`

E. `p->trapframe->ra`

F. `p->trapframe->epc`

Name:

[Handwritten signature]

Ben is using xv6 on a computer with just a single CPU (i.e. a single RISC-V hart). He thinks that, because threads cannot ever run truly in parallel with only one CPU, that therefore `acquire()` and `release()` are not needed. Ben modifies his kernel's `acquire()` and `release()` functions to simply return without doing anything, and he also eliminates the three checks and panics in `sched()` (since they are checking things to do with locks).

Ben has overlooked something important about the effects of xv6's spinlocks even when there is just one CPU!

11. [5 points]: Explain briefly why Ben's modifications will result in a broken kernel even though he only runs it on machines with a single CPU.

One side effect of `acquire()` is the intr will be disabled until the lock is released.

By removing that from `acquire()`, Ben made it possible for intr to occur during

a context switch, which can cause disasters. Think: what would happen, half way of reading registers from `c->ctx` in `switch()`, when a timer interrupt occurs?

Name:

F. H.

VI Copy-on-write fork

Ben's solution for COW-fork marks in `fork` all the pages below `p->sz` in the parent and child as read only. On a page fault, Ben's solution checks if the faulting address is below `p->sz`, and, if so, allocates a new page, copies the faulting page into the new page, and inserts the new page with read-write permissions in the process's page table. Ben's solution passes `cowtest`.

12. [5 points]: Briefly explain why Ben's solution is incorrect despite passing `cowtest`.

Beyond `p->sz`, there are trapframe that stores regs and trampoline

that is not writeable and mapped for all proc to the same physical page. Since ben simply copies the faulting page and make the new page RW, then someone could modify the trapframe, which could be a disaster.

I thought it meant beyond

Name:

SH

VII 6.1810

13. [1 points]: Which parts of the xv6 textbook were most helpful? Which parts did you think were most confusing?

Most helpful: comprehensive description, and real world cases.

Most confusing: it would be better to cite which location an instruction is in the RISC-V manual, or alternatively mention its whole function.

14. [1 points]: Please indicate which of the labs you found to be the most helpful in learning the material, and which the least.

A. Utilities

B. Sys calls Best

C. Page tables

D. Traps

Name:

E. Copy-on-write fork Worst: just one task

- 15. [1 points]:** What's the most important thing we could fix about 6.1810 to make it better?
Make the lecture videos open

End of Quiz I

Name:

F. S.